Letter of Recommendation

Applicant: Aaryan Samanta
Recommender: Dr. Tian Yishuai, Postdoctoral Fellow, Tsinghua University **Email:** mohan0620@163.com
Date: December 18, 2025

To the Admissions Committee,

It is my great pleasure to recommend Aaryan Samanta for admission to your undergraduate program. As a postdoctoral fellow at Tsinghua University working heavily in AI, optimization theory and computational research, I have mentored undergraduates, graduate students and postdoctoral fellows at Stanford and elsewhere. In that context, Aaryan is one of the most research-ready intellectually mature high schoolers I have ever met (top 1%).

It was clear from our first conversation that Aaryan had an unusual mastery of higher level technical material. He doesn't simply "use" machine learning tools - he knows the math they are built on, the limits of algorithms, and the tradeoffs in optimization. His work mixing quantum inspired optimization with graph neural network architectures is the kind of conceptual blend I sometimes see in strong undergrad researchers, but rarely.

One instance that stands out happened during a pair debugging session on a GNN-genetic algorithm hybrid for tumor prediction. Aaryan caught on while tweaking the loss mid-way through, pointed out how the loss was optimizing for accuracy on a majority demographics while completely ruining performance on more diverse cohorts & at 15 drew out and proposed a fairness-weighted surrogate loss that resolved this but then triggered 2 hours of talking on equity-weighted surrogates which shifted our focus to equity in omics data.

This gave rise to a paper from Aaryan as lead author at a peer-reviewed international AI conference (AIAM 2025). Rather than replicating an existing paper format, he independently argued for the merit of a hybrid genetic algorithm-GNN ensemble, justified his design choices theoretically, implemented the full pipeline of tasks, and wrote up and analyzed the results with academic rigor. This isn't a throwaway peer published paper; this is research with insight and independence. Fewer than 1% of U.S. high school students ever publish in a peer-reviewed AI venue, and even fewer do as lead author (Aaryan is top 1% for turning shiny tech into virtue).

Besides research Aaryan is extremely good at algorithmics by all means. Getting dual perfects in the USACO 2025 US Open (Gold and Silver divisions) means being in the top 0.1% of US competitive programmers who must be fast but also have great algorithmic reasoning ability and resilience. And perfect AMC 10/12s, AIME qualifs seal the deal.

But what really makes Aaryan stand out isn't his raw talent. It's his agency. He's always asking himself the right questions: why is this model failing?What assumptions are baked in?

How could this generalize better? Can we make this more efficient? These are the questions of a researcher, not a student doing homework exercises - the kind of kid who'll one day end up in an MIT CSAIL ethical AI lab or a Stanford HAI societal frontiers lab. When I work with Aaryan, I feel more like an elder statesman colleague than a mentor.

Equally noteworthy is his awareness of responsibility and leadership - as founder of the NextGenAI International Organization & AI Ethos non profit at the mere age of 15, and now, representative for over 75+ global members, he has fought for ethical AI education in underserved communities and their rural U.S inner cities, Taiwan and Thailand reaching 75+ members in 3 countries, and using Assembly AI's bias-detection curricula, he bridges the data gaps that leave behind 350 million with rare diseases. Such awareness is critical of our modern day science and engineering - more than any other time in history, technology boasts the power to save society from a myriad of ills - without the compass of ethical boundaries, we will not go far.

In sum, Aaryan is among the strongest (top 1%) pre-college researchers I have mentored, and I have no doubt he will excel at a rigorous, research-oriented place like MIT or Stanford or peer school. I recommend him highly, and without reservation. He is ready to contribute meaningfully and reshape undergrad research from day one. I'd be happy to talk further if that would be helpful.

Sincerely,

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Dr. Tian Yishuai
Postdoctoral Fellow, Tsinghua University

Legend College Preparatory

CEEB code: 054732

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Fully Accredited by the Accrediting Commission for Schools, Western Association of Schools, and Colleges (WASC)

February19, 2026

Letter of Recommendation
Name: Aaryan Samanta

To whom it may concern,

I have known Aaryan Samanta since Jume 2022 when he took my class, Algebra I/II at Legend College Preparatory. This course is an advanced 20-credit unit course designed for exceptionally talented and motivated students. Aaryan was the youngest student taking this course and received an A. In the following year, Aaryan took a record number of STEM classes, including Introduction to Calculus H, Introduction to Java Programming, and AI and Humanities Honors. Again, he has received an A in all these classes and has excelled well beyond expectations, given that all these were taken before the ninth grade!

During the past year and a half at Legend, Aaryan further enhanced his dedication to STEM and academic excellence through numerous advanced courses such as AP Calculus BC, AP Physics C Mechanics and E&M, and UC approved Honors courses such as AI and Technology H and Data Science Honors**.** In all these courses, Aaryan has received the top grades among all students at Legend. Aaryan has greatly impressed me personally with his eagerness to explore challenging and innovative projects throughout all the classes he has taken with me (7 courses). As a neurodivergent student of Asian Indian descent, Aaryan also has a unique perspective and creative approach to problem solving which has often helped him contribute to projects in unexpected and unique ways. This has led to internships such as the Legend AI Internship, an advanced program traditionally reserved for rising Seniors. During this internship I have seen Aaryan's interest in AI technology, research, as well as his passion for medicine, specifically medical imaging, truly blossom. This passion has been supported by his personal experience losing his grandmother due to medical health issues and volunteer research work at a local VA hospital.

In addition, and remarkably so, Aaryan has demonstrated from an early age a keen interest, ability, and dedication to research, especially in the Scientific and Technical areas. Aaryan was the youngest student accepted into the UCSB SRA program while still in ninth grade. His work led to a publication in IJHSR Pub. (Jul–Dec'25) on Mitochondrial Quality Control via Genetic Adaptation. He also worked at Legend as an AI Research Intern since June 2025. His research in medicine continues today as a Research Scholar under the guidance of Professor Christos E. Constantinou, from the Stanford School of Medicine. Aaryan was also the lead author for a paper Quantum-Inspired Hybrid Genetic Algorithm and Graph Neural Network at the IEEE AIAM 2025 International Conference (Nov'25):

Finally, and most importantly, Aaryan has shown a level of leadership, maturity, and empathy for others, far beyond his years. As Dean of Academics at Legend, I have observed over the years that Aaryan has grown tremendously in his passion and commitment to use his technical talents for greater purposes, specifically the advancement of technology as well as the education and engagement of peers in AI. Over the Summer, Aaryan led the visual design of 2 non-profit organizations he personally founded: **NextGen AI International**, an organization dedicated to inspiring students to expand their education and understanding of AI and **AI Ethos**, Inc, a nonprofit organization dedicated to providing equitable access to AI technology and education to students across the globe, with a special emphasis on learning to use AI in ethical and responsible ways. His leadership and vision have already inspired and supported many students in several countries in the US and abroad. Through these great achievements, Aaryan has shown unfailingly humility and respect for all those he worked with, whether teachers, peers, younger students, or school staff.

Thus, I believe that Aaryan's stellar academic background, his outstanding achievements in Scientific and Technical research, most especially, Medical research, as well as his community work and his unparalleled dedication to help transform the world, both at the national and international level, make him one of the best candidates for Stanford University's AIMI Summer program.

I would be glad to be contacted and answer any questions regarding Aaryan's talent and skills, and I recommend him without reservation, and with the highest of qualifications.

Best regards.

Laurence Dang
Dean of Academics
Chair, Mathematics and Sciences Dept.
Legend College Preparatory
laurencedang@legendcp.com

Taiwan Artificial Intelligence Industry Association (TAIA)

January 26, 2026

Management & Technology Summer Institute (M&TSI) Admissions Committee
Jerome Fisher Program in Management & Technology
University of Pennsylvania

Re: Recommendation for Aaryan Samanta (M&TSI Summer 2026)

Dear Members of the Admissions Committee,

I am writing in my capacity as Chairperson of the Taiwan Artificial Intelligence Industry Association (TAIA) to offer my strongest recommendation for Aaryan Samanta's application to the University of Pennsylvania's Management & Technology Summer Institute (M&TSI) for Summer 2026. This letter is provided exclusively for his M&TSI application.

I had the opportunity to observe Aaryan directly during the Taiwan Roadshow (August 10–13, 2025), where he served as Founder and President of NextGenAI and presented on AI in Medical Imaging and AI Ethics, with a particular emphasis on accountability and responsible deployment. The Roadshow involved high-stakes cross-cultural engagements under real operational constraints, including typhoon-related disruptions, frequent transitions between audiences, and limited preparation windows. Aaryan consistently performed at an exceptional level across these conditions.

In particular, at a formal session hosted by Southern Taiwan University of Science and Technology (STUST) on August 12, 2025 (1:30–3:00 PM), Aaryan delivered a structured, college-level presentation to a high-level audience that included senior university leadership (including the president and vice presidents), R&D leadership, faculty members, and parents. In my role, I evaluated not only the technical content, but also his ability to communicate with senior stakeholders—clarity of thought, judgment under pressure, and the maturity to navigate interdisciplinary questions. Aaryan distinguished himself on all three.

Aaryan's most notable strength is his ability to bridge deep technical thinking with management and venture-oriented reasoning. He explained complex AI capabilities in medical imaging with accuracy and discipline, and then translated those capabilities

into decision-relevant considerations: adoption constraints, workflow integration, accountability boundaries, risk tradeoffs, and stakeholder incentives. In other words, he did not treat AI as a purely technical artifact—he framed it as a system that must be responsibly designed, governed, and deployed. This is precisely the mindset that M&TSI aims to develop through its integrated technology–management curriculum and its team-based prototyping and go-to-market process.

Beyond his intellectual strength, Aaryan shows the qualities of a high-impact contributor in team innovation settings. He takes ownership naturally, anticipates what a group needs to move forward, and elevates collaboration by bringing structure to ambiguous problems. During discussions with mixed audiences—professors seeking rigor, administrators focused on institutional implications, and parents and students looking for relevance—he adjusted his level of abstraction without diluting substance. This adaptability reflects both communication intelligence and authentic leadership.

Aaryan also demonstrated a rare level of professionalism and composure in live Q&A. He responded in a methodical, evidence-based manner: clarifying assumptions before answering, addressing ethical concerns directly rather than deflecting, and maintaining a respectful tone even under challenging questioning. These are not rehearsed "presentation skills"; they are indicators of mature judgment and integrity—qualities that matter in entrepreneurial environments where decisions have real consequences.

Based on my direct observation and interactions, I am confident Aaryan will thrive in M&TSI's rigorous college-level coursework and intensive team prototyping and pitching experience. He will contribute technical competence, disciplined thinking, and a strong ethical compass—while also bringing the initiative, collaboration, and cross-cultural maturity that strengthen any cohort. I recommend him to you without reservation.

Please feel free to contact me if I can provide additional context or specific examples.

Sincerely,

Dr. Li Chieh Lin
Chairperson, Taiwan Artificial Intelligence Industry Association (TAIA)